PES UNIVERSITY

SE LAB -06

Lab 6: Code and Branch Coverage Analysis

PROJECT TITLE: L6_L7_61_credit_card_rewards

NAME: PRAJWAL G T

SRN: PES2UG25CS818

SECTION:H

1. Initial Code and Test Case Analysis

1.1 Application Code

The application file credit_card_rewards.py contains three functions: reward_points(), late_payment_fee(), and card_status(). These functions implement credit card reward point calculations, late payment fee calculations, and card status determination.

The application code was tested using the existing test file, test_credit_card_rewards.py.

BEFORE THE CODE CHANGE HAVE THE ERROR

```python
POINTS_PER_100 = {"classic": 1, "gold": 2, "platinum": 4}


def reward_points(card_tier, amount, category, is_weekend):
    if card_tier not in POINTS_PER_100:
        raise ValueError("unknown card tier")
    if amount <= 0:
        raise ValueError("invalid amount")
    if category == "fuel":
        return 0
    points = amount // 100 * POINTS_PER_100[card_tier]
    if category == "dining" or category == "travel":
        points = points * 2
    if is_weekend and card_tier != "classic":
        points += 50
    return points


def late_payment_fee(outstanding, days_late):
    if days_late <= 0 or outstanding <= 0:
        return 0
    if outstanding <= 1000:
        fee = 100
    elif outstanding <= 10000:
        fee = 500
    else:
        fee = 1000
    if days_late > 30:
        fee += 750
    return fee
```

```python
def late_payment_fee(outstanding, days_late):
    if days_late <= 0 or outstanding <= 0:
        return 0
    if outstanding <= 1000:
        fee = 100
    elif outstanding <= 10000:
        fee = 500
    else:
        fee = 1000
    if days_late > 30:
        fee += 750
    return fee


def card_status(utilisation_percent, missed_payments, is_frozen):
    if is_frozen:
        return "frozen"
    if missed_payments >= 3:
        return "blocked"
    if utilisation_percent > 90:
        return "over limit risk"
    elif utilisation_percent > 30 or missed_payments > 0:
        return "watch"
    return "good"
```

1.2 Initial Test Cases

Initially, the test file contained two test cases:

1. test_classic_grocery() – checks whether a classic card earns the correct reward points for a grocery transaction.
2. test_paid_on_time() – checks whether the late payment fee is zero when the payment is made on time.

These tests passed successfully, but they did not cover all the conditions and branches present in the application code.

```python
import pytest
from credit_card_rewards import reward_points, late_payment_fee, card_status

#Example test cases:
def test_classic_grocery():
    assert reward_points("classic", 1000, "grocery", False) == 10

def test_paid_on_time():
    assert late_payment_fee(5000, 0) == 0

#Write the rest of the test cases so that the code is 90- 100% covered.
```

1.3 Initial Coverage Results

The initial coverage analysis showed:

- **Code coverage:** 37%
- **Branch coverage:** 31%
- **Test cases passed:** 2 out of 2

The results indicated that several conditions, decision branches, and edge cases were not tested. Additional test cases were therefore required to improve code coverage and branch coverage to at least 90%, as specified in the lab requirements.

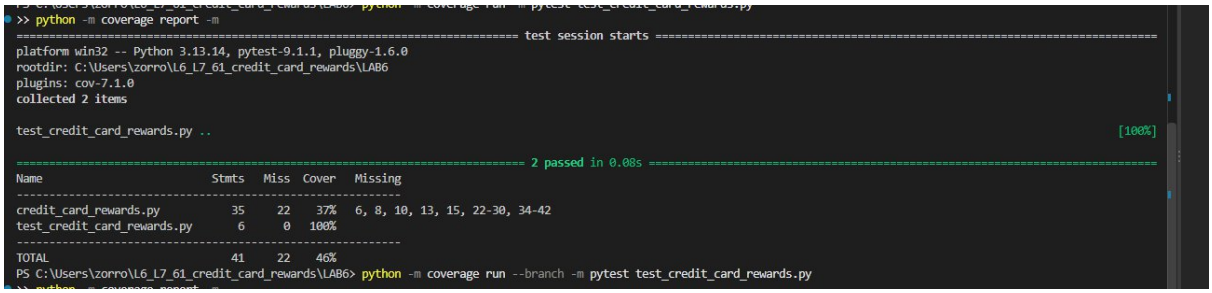
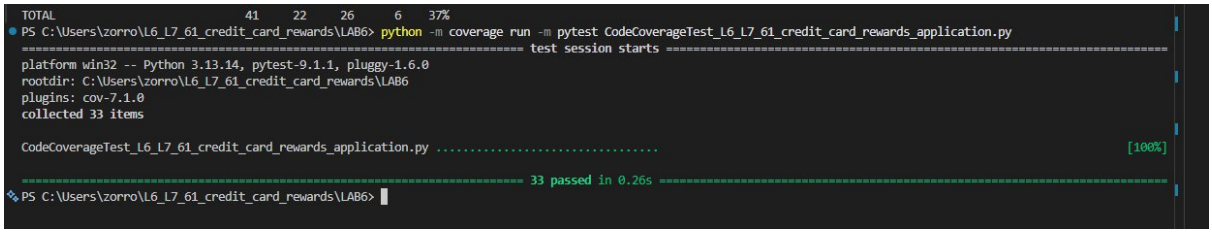


Figure: Code coverage analysis after adding test cases. All 33 tests passed successfully, and the application achieved 100% code coverage.



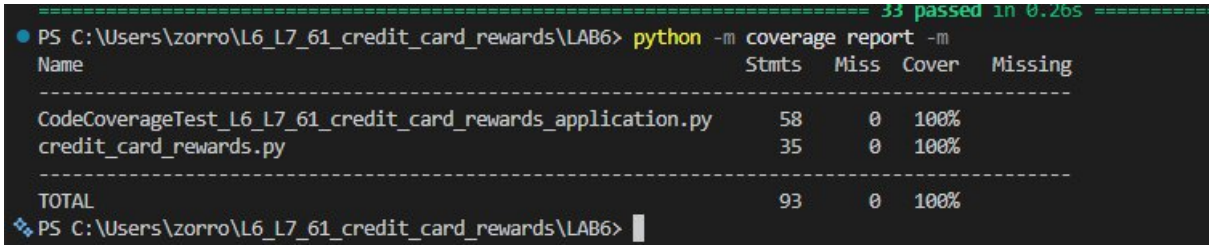
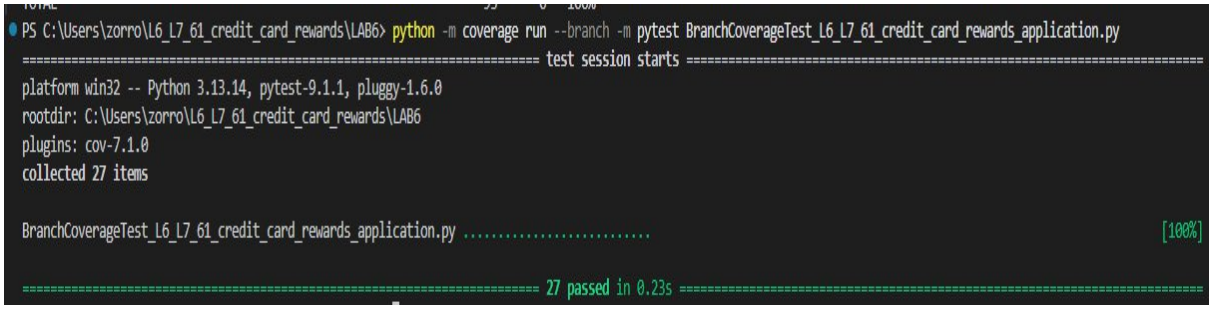
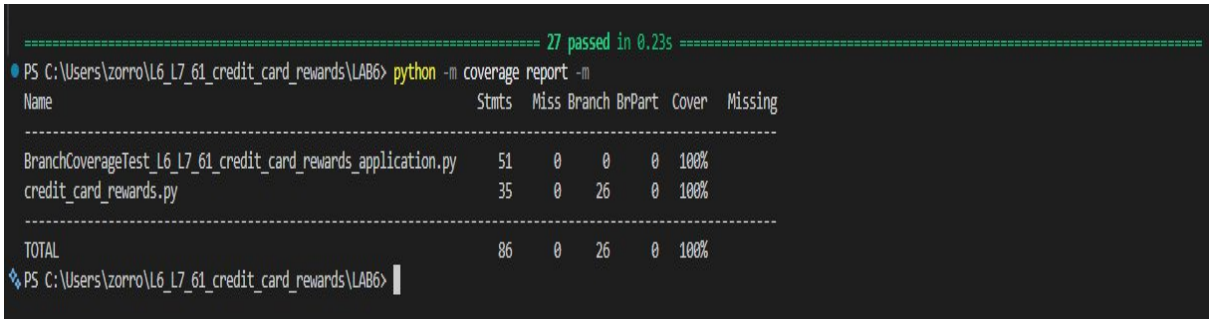


Figure: Branch coverage analysis after adding test cases. All 27 tests passed successfully, and the application achieved 100% branch coverage with all 26 branches covered.





5. Before-and-After Summary

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Note: The initial values are from the initial report shown in the terminal screenshot. The final values are from the subsequent test runs shown in the provided screenshots.

6. Conclusion

Additional tests were added without changing the application logic. The code coverage test suite passed 33 tests and reported 100% coverage. The branch coverage test suite passed 27 tests and reported 100% coverage, with all 26 branches covered. These results exceed the lab requirement of at least 90% coverage.